Qns	Answer	Marks
1(a)(i)	gravitational force of attraction per unit mass acting on (OR experienced by) a small test mass placed at that point (in the gravitational field)	**1**
1(a)(ii)	gravitational force of attraction between **two point masses** is directly proportional to the product of the masses and inversely proportional to the **square of separation between the masses** Let *m* be the mass of an object distance *R* away from Mass *M* field strength is gravitational force of attraction per unit mass experienced by **small test mass** placed at that point $\dfrac{F}{m} = \dfrac{1}{\cancel{m}}\dfrac{GM\cancel{m}}{r^2} = \dfrac{GM}{r^2}$	**1** **1** **1**
1(b)(i)	$\text{volume of star} = \dfrac{4}{3}\pi r^3$ $= \dfrac{4}{3}\pi\left(2.7\times10^4\right)^3$ $= 8.245\times10^{13}\ \text{m}^3$ $\rho = \dfrac{m}{V} = \dfrac{6.2\times10^{30}}{\frac{4}{3}\pi\left(2.7\times10^4\right)^3} = 7.52\times10^{16}\ \text{kg m}^{-3}$	**1** **1**
1(b)(ii)	(words to effect of) density increase closer to centre (words to effect of) outer layers compress inner layers	**1** **1**

Qns	Answer	Marks
1(b)(iii)	By conservation of energy: loss in KE $=$ gain in GPE $\frac{1}{2}mv^2 - 0 = 0 - \left(-\frac{GMm}{r}\right)$ $v = \sqrt{\frac{2Gm}{r}} = \sqrt{\frac{2\left(6.67\times10^{-11}\right)\left(6.2\times10^{30}\right)}{2.7\times10^{4}}}$ $= 1.75\times10^{8}\ \text{m s}^{-1}$	 1 1

Qns	Answer	Marks
2(a)	oscillatory motion where	